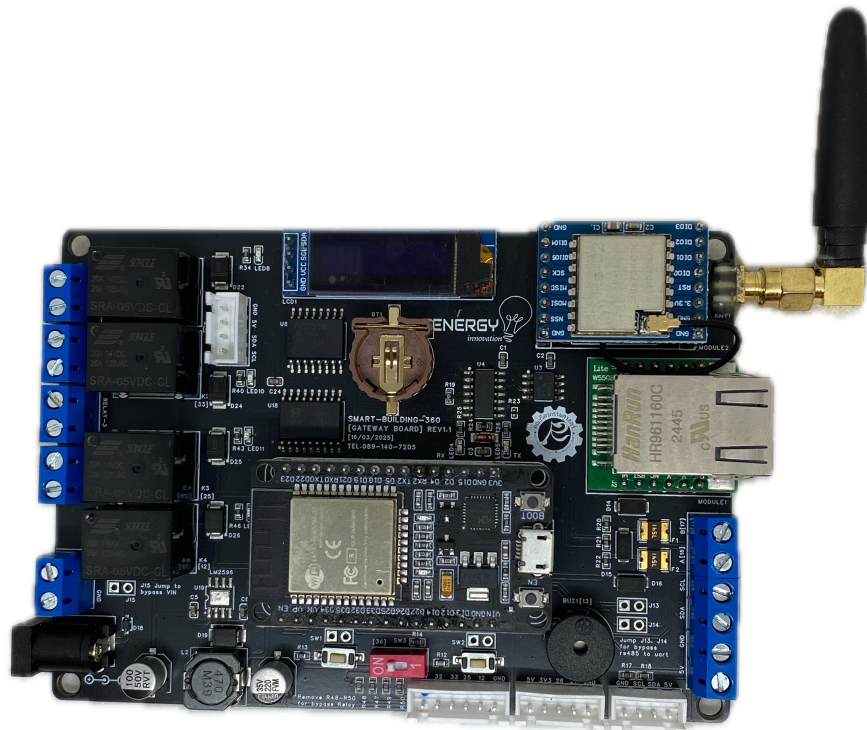


Tenergy32Gateway – ESP32-Based IoT Gateway

Model: Tenergy32Gateway

Designed & Manufactured by: TENERGY INNOVATION CO., LTD.

📍 www.tenergyinnovation.co.th | ☎ +66 89-140-7205



Product Overview

The Tenergy32Gateway is a powerful and industrial-grade ESP32-based IoT gateway board designed for smart building systems, automation, and sensor networks. It features long-range LoRa communication, Ethernet connectivity, RS485 support, and multiple interfaces for real-world deployment in both standalone and cloud-integrated IoT environments.

Technical Specifications

- Microcontroller: ESP32-DOIT-DEVKIT-V1 (dual-core, Wi-Fi + BLE)
- Power Supply: Supports wide-range DC input from 4.5V to 60V
- Display: Integrated 128x32 OLED display for real-time data/status
- LoRa Module: SLX1278 operating at 433 MHz with external SMA antenna
- Ethernet Module: W5500 Ethernet chip with RJ45 port for LAN connection
- User Interface: 2x tactile switches and 1x slide switch for configuration or input
- Buzzer: 2.048 kHz onboard piezo buzzer for audible notifications
- Relay Output: 4x Relay modules (COM, NO terminals) for AC/DC load control

- • RS-485 Port: Industrial-grade RS485 port compatible with PLCs, power meters, sensors, HVAC, inverters
- • I2C Interface: For external sensors (SHT30, BH1750, MPU6050, BMP280, etc.)
- • UART / RS232 Interface: Compatible with NEO-6M, MH-Z19B, RC522, EM18, barcode scanners, RS232 converters
- • Mesh Networking: Supports Wi-Fi Mesh, ESP-NOW, and LoRa-based Mesh
- • RTC Module: Onboard DS3231 Real-Time Clock with battery backup
- • Digital I/O: Configurable GPIOs for digital input/output

Specifications Overview

Feature	Description
Microcontroller	ESP32-DOIT-DEVKIT-V1, Dual Core 240MHz, Wi-Fi + BLE
Operating Voltage	DC 4.5V to 60V wide input
Display	OLED 128x32 pixels for status or feedback
LoRa Module	SLX1278, 433 MHz with external antenna
Ethernet Module	W5500 Ethernet with RJ45 port
Switches	2x Tactile Push Buttons, 1x Slide Switch
Buzzer	2.048 kHz Piezoelectric Buzzer
Relay Outputs	4x Relay modules (COM, NO terminals)
RS485 Port	For Modbus RTU communication with industrial devices
I2C Interface	Supports sensors like SHT30, BH1750, MPU6050, BMP280
UART/RS232 Port	For GPS, RFID, CO ₂ , RS232 devices
RTC	DS3231 Real-Time Clock with battery backup
Digital I/O	Configurable GPIOs
Mesh Networking	LoRa Mesh, Wi-Fi Mesh, ESP-NOW

Applications

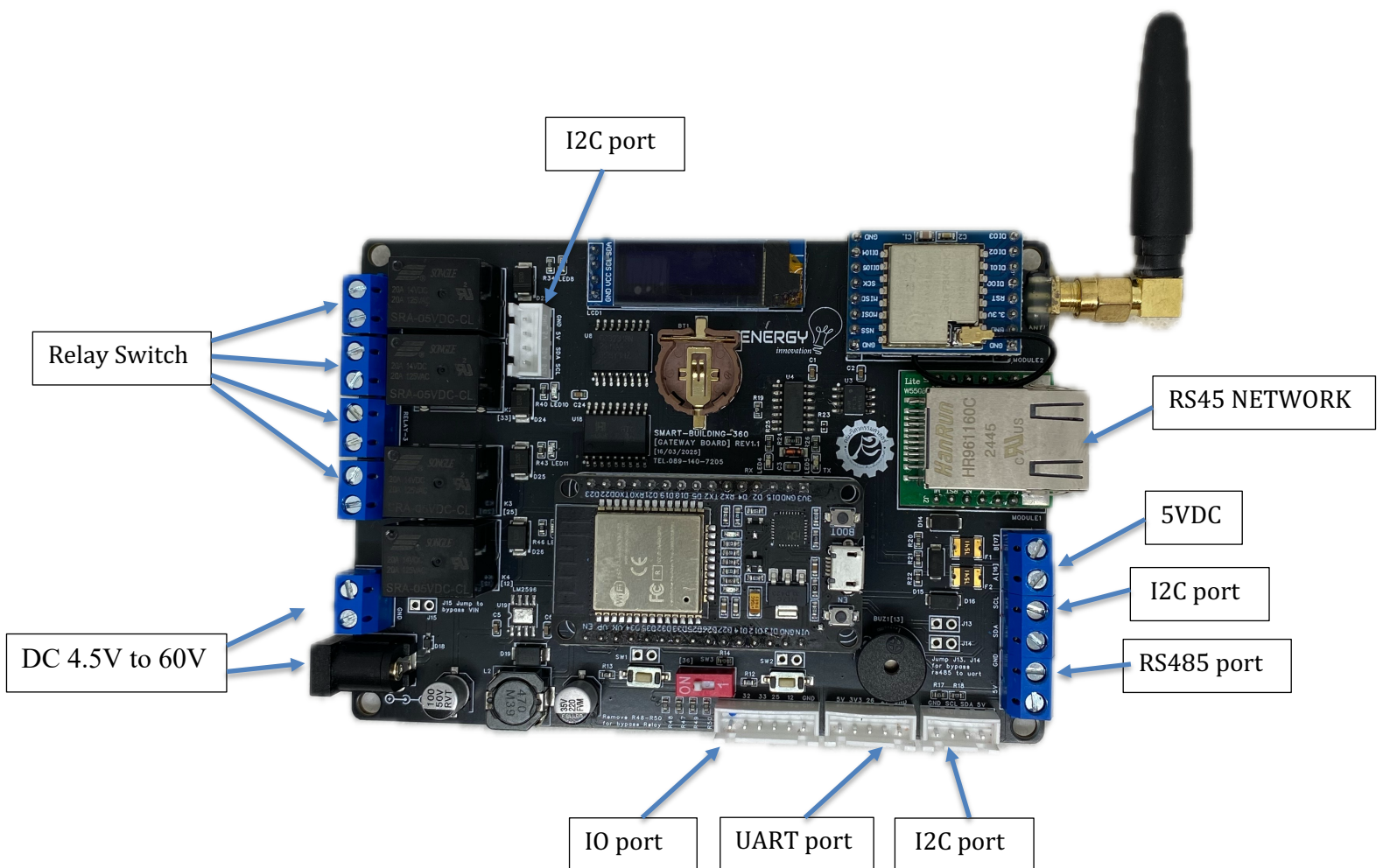
- • Smart Building Control Systems
- • Industrial Automation Gateways
- • LoRa Data Aggregators
- • Smart Energy Monitoring
- • IoT Cloud-Connected Controllers

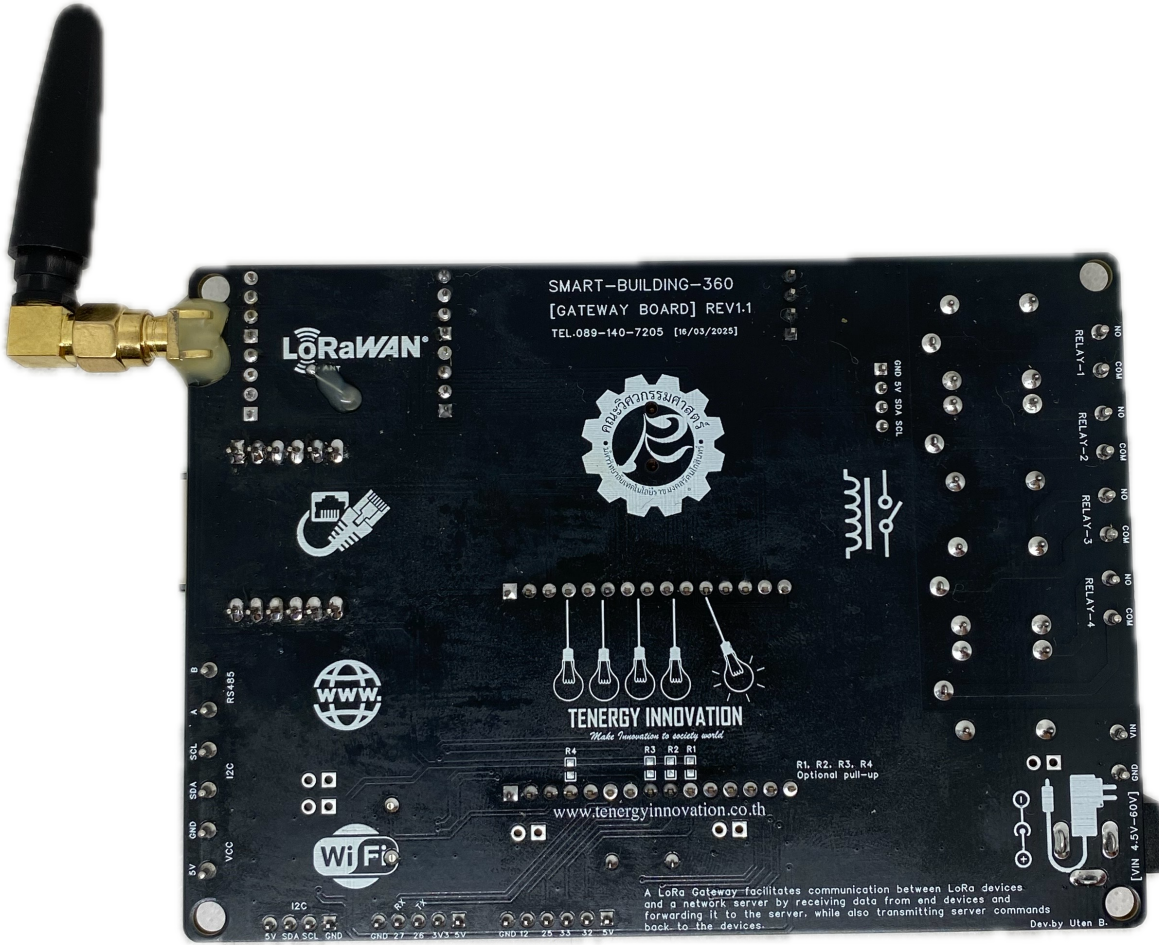
Board Dimensions & Mounting

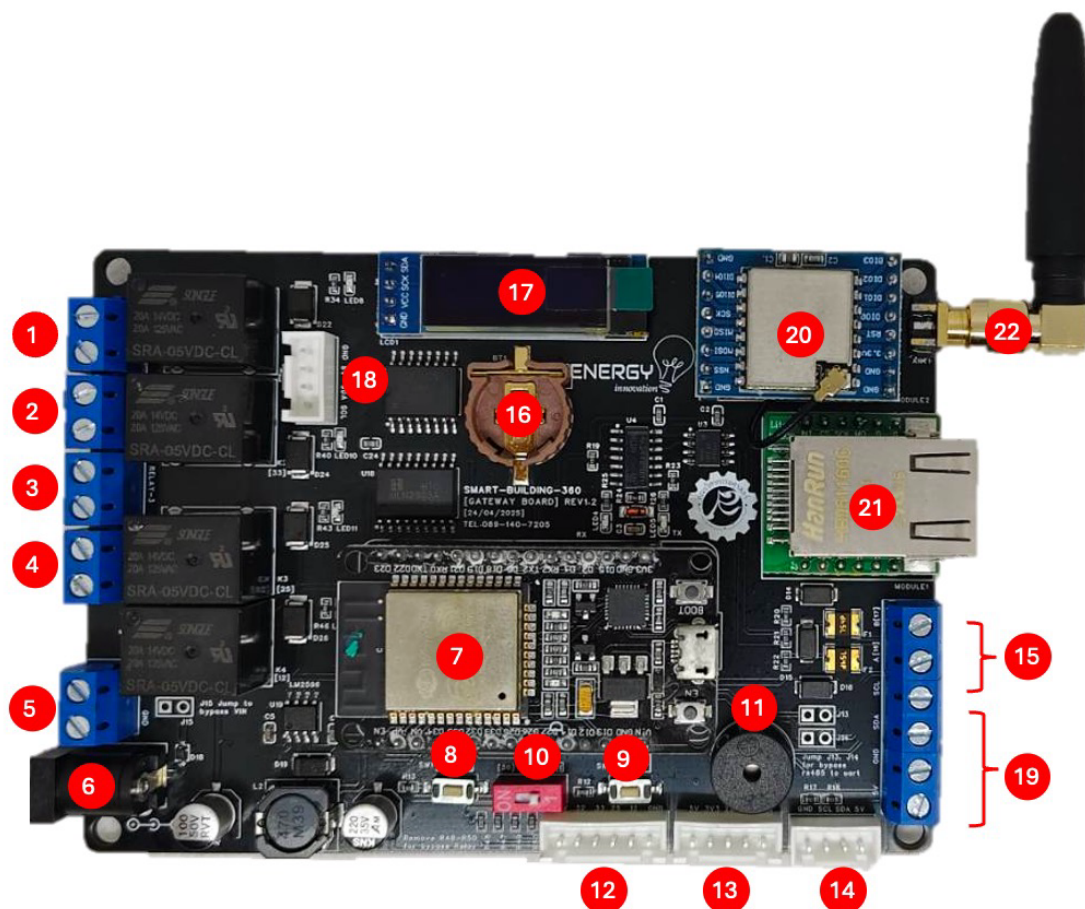
- Dimensions: ~120mm x 80mm
- Mounting: 4x M3 screw holes
- Antenna: SMA connector with external LoRa antenna

Development Support

- Programming Interface: USB (via onboard ESP32 module)
- Firmware Platforms: Arduino IDE / PlatformIO
- OTA Support: Wi-Fi Over-the-Air (OTA) updates supported
- Example Libraries: LoRa, OLED, Ethernet, RS485, I2C, UART, Mesh







Number	Description	Note
1	Relay1	Switch for controlling power circuit; connected to ESP32 pin 32 via COM and NO terminals on the left terminal block
2	Relay2	Switch for controlling power circuit; connected to ESP32 pin 33 via COM and NO terminals on the left terminal block

3	Relay3	Switch for controlling power circuit; connected to ESP32 pin 25 via COM and NO terminals on the left terminal block
4	Relay4	Switch for controlling power circuit; connected to ESP32 pin 12 via COM and NO terminals on the left terminal block
5	DC-IN Terminal	12VDC power terminal to supply board power
6	DC Power Jack	12VDC adapter jack, directly connected to terminal 5
7	ESP32- DOIT- DEVKIT-V1	Main microcontroller
8-9	Tactile Switches	Push-button switches SW1, SW2 on pins 34 and 35 for programming and testing
10	DIP Switches	Slide switch SW3 on pin 36 for programming and testing
Number	Description	Note
12	GPIO Connector	Connector for ESP32 pins 32, 33, 25, 12 as digital I/O

13	UART Connector	UART communication connector on ESP32 pins 26 and 27 (can be used as digital input)
14	I2C Connector	I2C connector using SDA and SCL lines for external I2C devices
15	RS485 Connector	RS485 module for long-distance, noise-resistant communication via ESP32 UART
16	Battery Backup	3VDC backup battery for DS3231 RTC to maintain time without main power
17	OLED Display	0.91" 128×32 pixel I2C OLED display for status and sensor data
18	I2C Connector	Additional I2C connector for external I2C devices on the same bus
19	I2C Connector	Supplementary I2C connector supporting multiple modules simultaneously
20	LoRa Module SLX1280	433MHz SLX1280 LoRa module via SPI for Sensor Board communication
21	RJ45 Ethernet Port	RJ45 port for LAN/Internet connection via W5500 Ethernet module
22	LoRa Antenna	Antenna connector for LoRa module to extend transmission range